Bounded Item Response Theory

Continuous Responses with Zero and One Inflation

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1. Introduction: Motivation and Scope

- Continuous item scores occur in visual analogue scales, sliders, proportions, and bounded response times.
- The theoretical bounds define the response scale.
- Johnson S_B, beta, and simplex distributions describe interior responses.
- Exact zeros and ones require explicit boundary probabilities (Molenaar et al. 2022).
- `BoundIRT` implements these models (Molenaar 2026).
- `BoundedIRT.Rmd` provides the derivations, code, and saved analyses.

1.1 Response Scale and Measurement Objective

For theoretical bounds $L < U$, rescale the original score:

$$Y_{ij} = \frac{Y'_{ij} - L}{U - L}, \qquad 0 \leq Y_{ij} \leq 1.$$

Relate the response from person i to item j to a latent trait:

$$\eta_{ij} = a_j \theta_i + b_j.$$

- θ_i: person position on the latent scale.
- $a_j > 0$: item discrimination.
- $b_j \in \mathbb{R}$: item easiness or intercept.
- The model specifies a full conditional response distribution.

2. Bounded Distributions: Johnson S_B

$$W \sim N(\mu, \delta), \qquad Y = \text{logit}^{-1}(W), \qquad \delta = \sigma^2 > 0.$$

$$f(y \mid \mu, \delta) = \frac{\exp[-\{\text{logit}(y) - \mu\}^2/(2\delta)]}{y(1-y)\sqrt{2\pi\delta}}, \qquad 0 < y < 1.$$

$$\text{Med}(Y) = \text{logit}^{-1}(\mu), \qquad F(y) = \Phi\left(\frac{\text{logit}(y) - \mu}{\sqrt{\delta}}\right).$$

- μ is a logit-scale location, rather than the response mean.
- The response mean and variance require numerical integration.
- δ is the variance on the logit scale (Johnson 1949).

2.1 Beta Distribution

$$f(y \mid c, d) = \frac{y^{c-1}(1-y)^{d-1}}{B(c,d)}, \quad 0 < y < 1, \quad c, d > 0.$$

Mean–precision parameterization (Ferrari and Cribari-Neto 2004):

$$\mu = \frac{c}{c+d}, \qquad \phi = c + d, \qquad c = \mu\phi, \quad d = (1-\mu)\phi.$$

$$E(Y) = \mu, \qquad \text{Var}(Y) = \frac{\mu(1-\mu)}{1+\phi}.$$

- Increasing precision ϕ reduces variance at a fixed mean.
- Shapes include uniform, skewed, J-shaped, and U-shaped densities.
- A continuous beta distribution assigns zero mass to exact endpoints.

2.2 Simplex Distribution

The simplex family was introduced by Barndorff-Nielsen and Jørgensen (1991).

$$f(y \mid \mu, \phi) = \{2\pi\phi[y(1-y)]^3\}^{-1/2} \exp\left[-\frac{(y-\mu)^2}{2\phi y(1-y)\mu^2(1-\mu)^2}\right].$$

$$0 < y < 1, \qquad 0 < \mu < 1, \quad \phi > 0, \qquad E(Y) = \mu.$$

- Here ϕ is a dispersion parameter.
- At small dispersion, $\text{Var}(Y) \approx \phi\mu^3(1-\mu)^3$.
- As $\phi \to \infty$, variance approaches $\mu(1-\mu)$.
- Use the exact variance for high-dispersion moment matching.

2.3 Parameter Interpretation Across Families

Family	Location	Additional parameter
Johnson S_B	Logit of median μ	Logit-scale variance δ
Beta	Response mean μ	Precision $\phi = c + d$
Simplex	Response mean μ	Dispersion ϕ

- Equal numerical parameter values have different meanings across families.
- Compare response moments and fitted densities on the original scale.
- All three families describe $0 < Y < 1$.

2.4 Variance Restrictions

For any response on $[0, 1]$, with mean m and variance v:

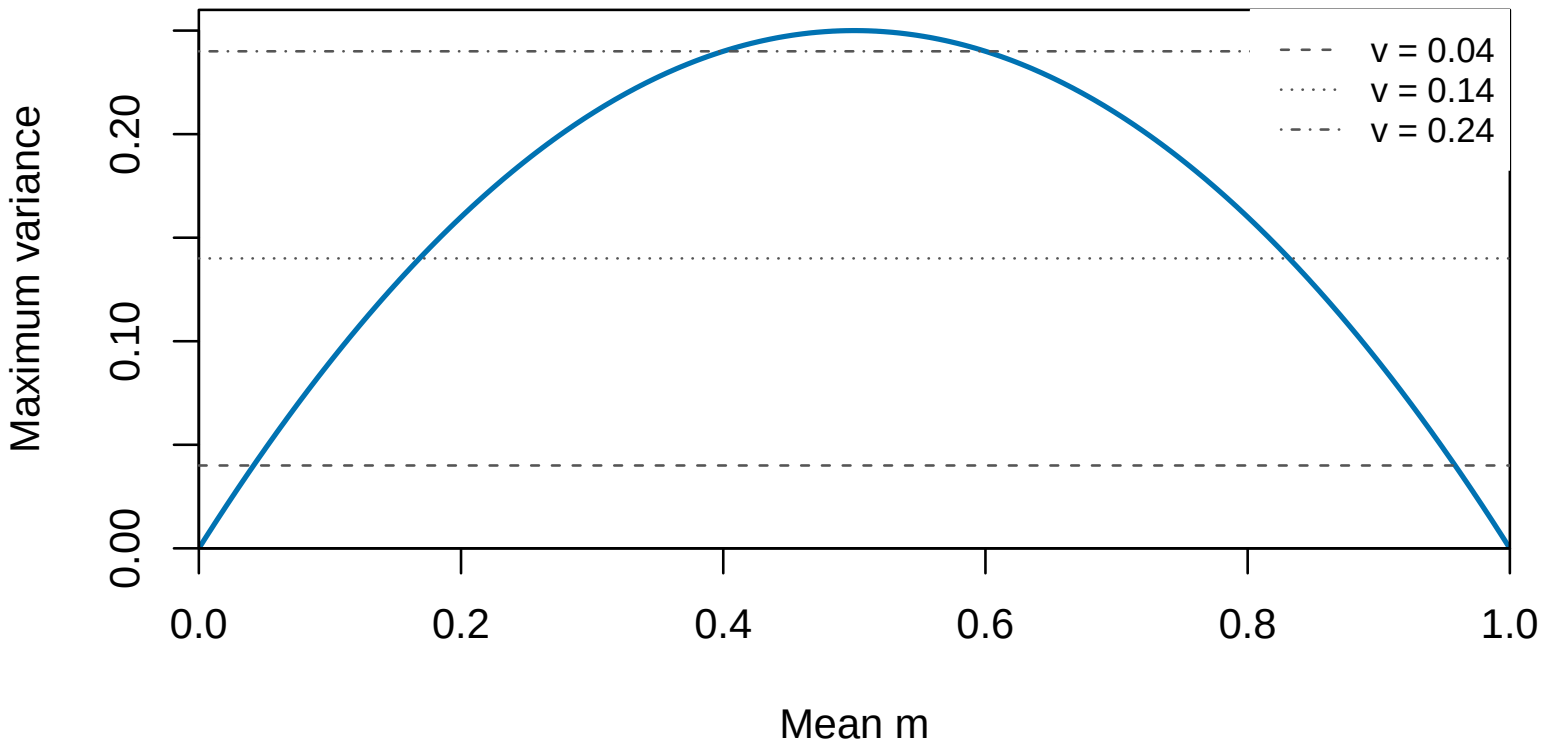
$$0 \leq v \leq m(1-m) \leq \frac{1}{4}.$$

Since $Y^2 \leq Y$,

$$v = E(Y^2) - m^2 \leq m - m^2.$$

- Equality $v = m(1-m)$ requires a Bernoulli distribution.
- Nondegenerate continuous responses satisfy $0 < v < m(1-m)$.
- A proposed mean–variance combination must satisfy this restriction.

2.5 Variance Envelope

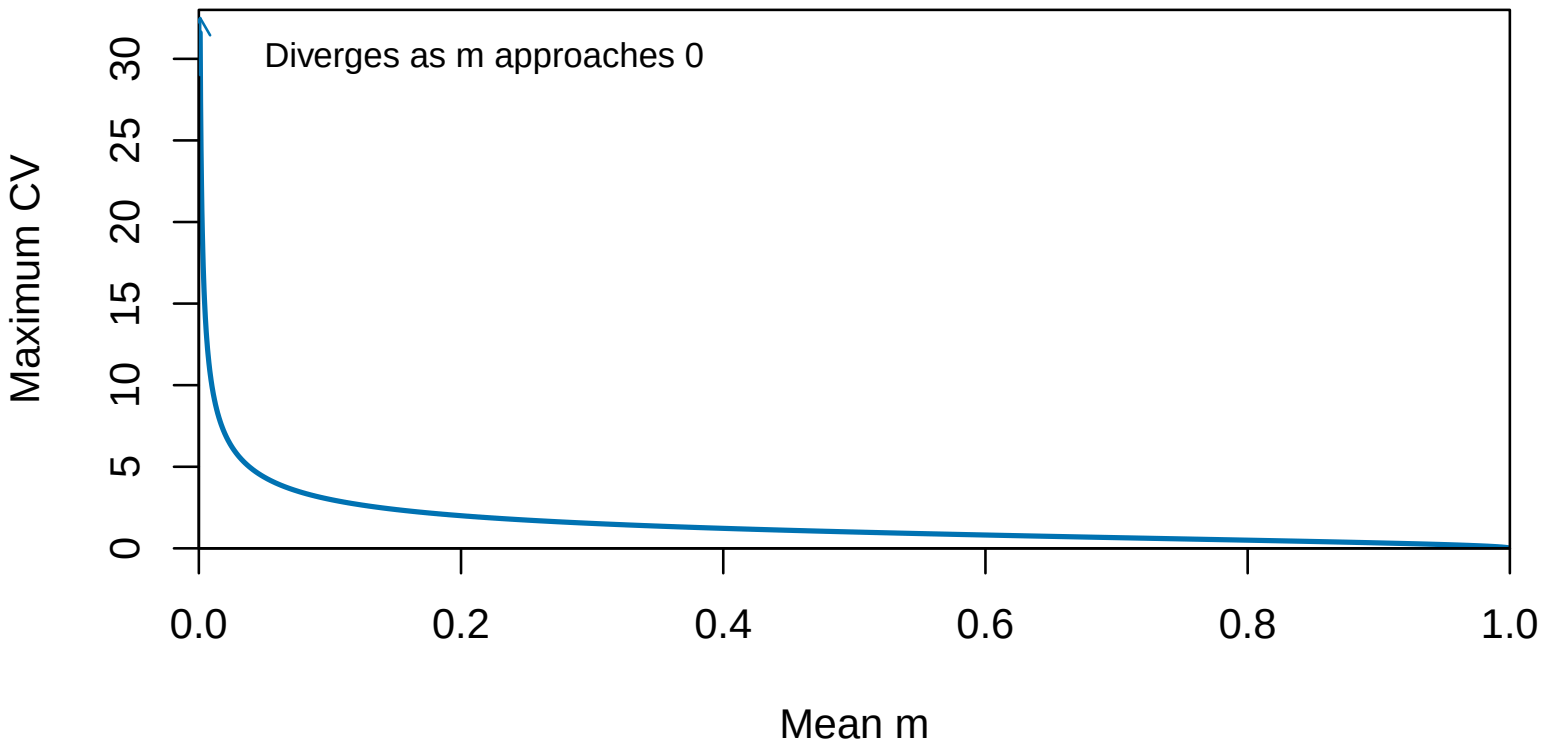


2.6 Coefficient of Variation

$$\mathrm{CV}(Y) = \frac{\sqrt{v}}{m}, \qquad 0 \leq \mathrm{CV}(Y) \leq \sqrt{\frac{1-m}{m}}.$$

- The bound applies for $0 < m \leq 1$.
- At $m = 0$, the CV is undefined.
- The maximum CV equals 2 at $m = 0.20$ and 1 at $m = 0.50$.
- Small means can produce large CVs even with small absolute variance.

2.7 CV Envelope



2.8 Moment Matching

Target means $m \in \{0.20, 0.50, 0.70\}$ and variances $v \in \{0.04, 0.14, 0.24\}$.

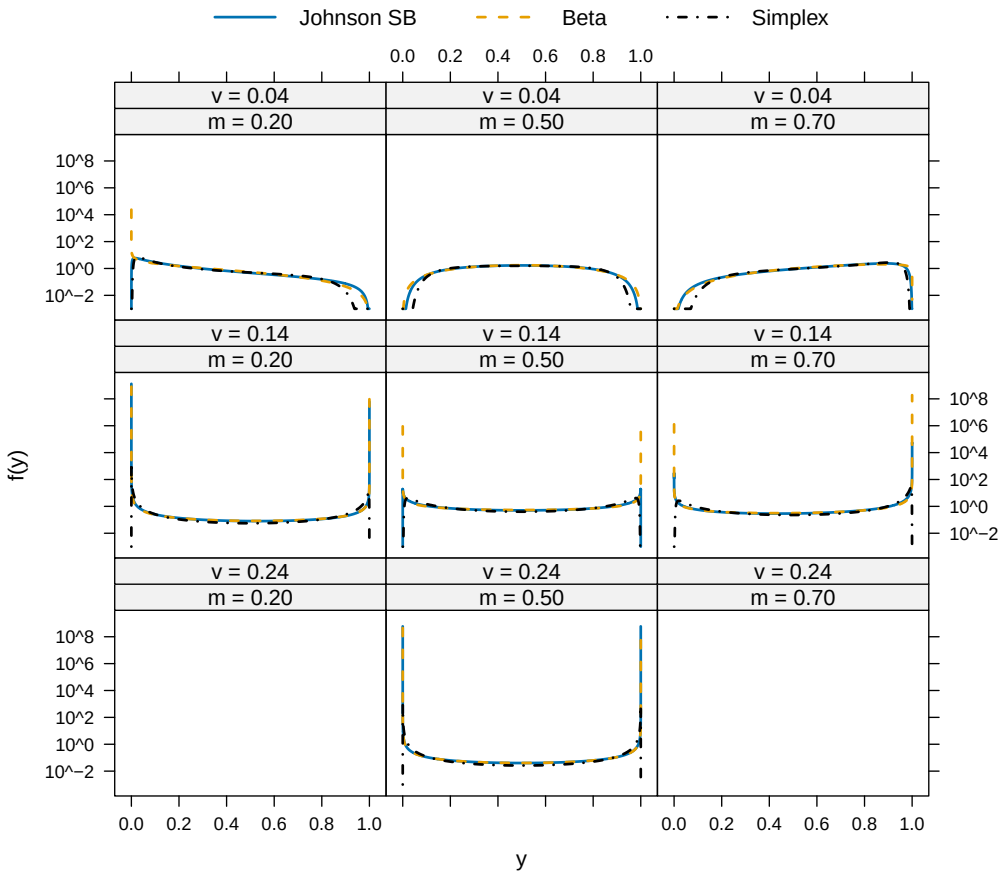
Mean	Maximum variance	Feasible target variances
0.20	0.16	0.04, 0.14
0.50	0.25	0.04, 0.14, 0.24
0.70	0.21	0.04, 0.14

Beta parameters follow directly:

$$\phi = \frac{m(1-m)}{v} - 1, \qquad c = m\phi, \quad d = (1-m)\phi.$$

Johnson S_B and simplex parameters require numerical matching.

2.9 Moment-Matched Density Shapes



Logarithmic density axis with display floor 10^{-3}. Empty panels are infeasible.

2.10 Identification and Monotonicity

- Johnson S_B inherits identification from the normal family through the invertible logit transformation.
- Beta mean and variance determine both shapes.
- The simplex mean and density identify its dispersion.
- Distributional identification still requires latent-scale constraints in IRT.
- Johnson S_B is stochastically increasing in μ at fixed δ.
- Beta is stochastically increasing in its mean at fixed precision.
- An increasing simplex mean alone does not prove stochastic ordering.

2.11 Skewness and Kurtosis

$$\text{Skew}(Y) = \frac{E[(Y-m)^3]}{v^{3/2}}, \qquad \text{Kurt}(Y) = \frac{E[(Y-m)^4]}{v^2}.$$

At matched $m = 0.20$ and $v = 0.04$:

Family	Skewness	Kurtosis
Johnson S_B	1.4049	4.4097
Beta	1.2000	3.8000
Simplex	1.3335	3.9179

Kurtosis here is the standardized fourth central moment, not excess kurtosis. Equal first two moments can conceal different response shapes.

3. Bounded Continuous IRT: General Structure

$$
\begin{aligned}
Y_{ij} \mid \theta_i, \boldsymbol{\tau}_j &\sim \mathcal{D}_j(\eta_{ij}, \zeta_j), \\
\eta_{ij} &= a_j \theta_i + b_j, \\
\theta_i &\sim N(0, 1).
\end{aligned}
$$

- Conditional local independence across items.
- Standard-normal person distribution fixes latent location and scale.
- Positive discrimination fixes the direction of the trait.
- Family-specific ζ_j controls conditional spread (Molenaar et al. 2022).

3.1 Easiness and Difficulty

$$\eta_{ij} = a_j\theta_i + b_j = a_j(\theta_i - b_j^*), \qquad b_j^* = -\frac{b_j}{a_j}.$$

- Larger easiness b_j shifts the response curve to the left.
- Larger difficulty b_j^* shifts it to the right.
- At $\theta_i = b_j^*$, the predictor equals zero.
- At that point, the beta and simplex interior means equal 0.5.
- The Johnson S_B interior median also equals 0.5.

3.2 Samejima or Johnson S_B IRT

$$\text{logit}(Y_{ij}) \mid \theta_i \sim N(a_j\theta_i + b_j, \delta_j).$$

$$\text{Med}(Y_{ij} \mid \theta_i) = \text{logit}^{-1}(a_j\theta_i + b_j).$$

- $\delta_j > 0$ denotes variance on the logit scale.
- The conditional mean generally differs from this logistic median.
- Without inflation, item information is constant:

$$I_j(\theta) = \frac{a_j^2}{\delta_j}.$$

(Samejima 1973; Molenaar et al. 2022)

3.3 Beta IRT

$$Y_{ij} \mid \theta_i \sim \text{Beta}(c_{ij}, d_{ij}),$$

$$c_{ij} = \exp\left(\frac{\eta_{ij} + \log \lambda_j}{2}\right), \qquad d_{ij} = \exp\left(\frac{-\eta_{ij} + \log \lambda_j}{2}\right).$$

$$E(Y_{ij} \mid \theta_i) = \text{logit}^{-1}(\eta_{ij}), \qquad c_{ij} d_{ij} = \lambda_j.$$

- The item parameter λ_j is the product of shapes.
- Precision varies across persons: $\phi_{ij} = 2\sqrt{\lambda_j}\cosh(\eta_{ij}/2)$.
- A fixed λ_j therefore does not imply fixed precision.

(Noel and Dauvier 2007; Molenaar et al. 2022)

3.4 Simplex IRT and Normal Benchmark

Simplex IRT model (Molenaar et al. 2022):

$$Y_{ij} \mid \theta_i \sim \text{Simplex}(\mu_{ij}, \phi_j), \qquad \mu_{ij} = \text{logit}^{-1}(\eta_{ij}).$$

- μ_{ij} is the conditional response mean.
- $\phi_j > 0$ controls dispersion.
- Information can increase toward trait extremes.

Normal benchmark (Mellenbergh 1994):

$$Y_{ij} \mid \theta_i \sim N(\eta_{ij}, \sigma_j^2).$$

The normal model assigns positive probability outside the bounded scale.

Flores et al. (2020) use the simplex distribution in a joint model for bounded response times and response accuracy.

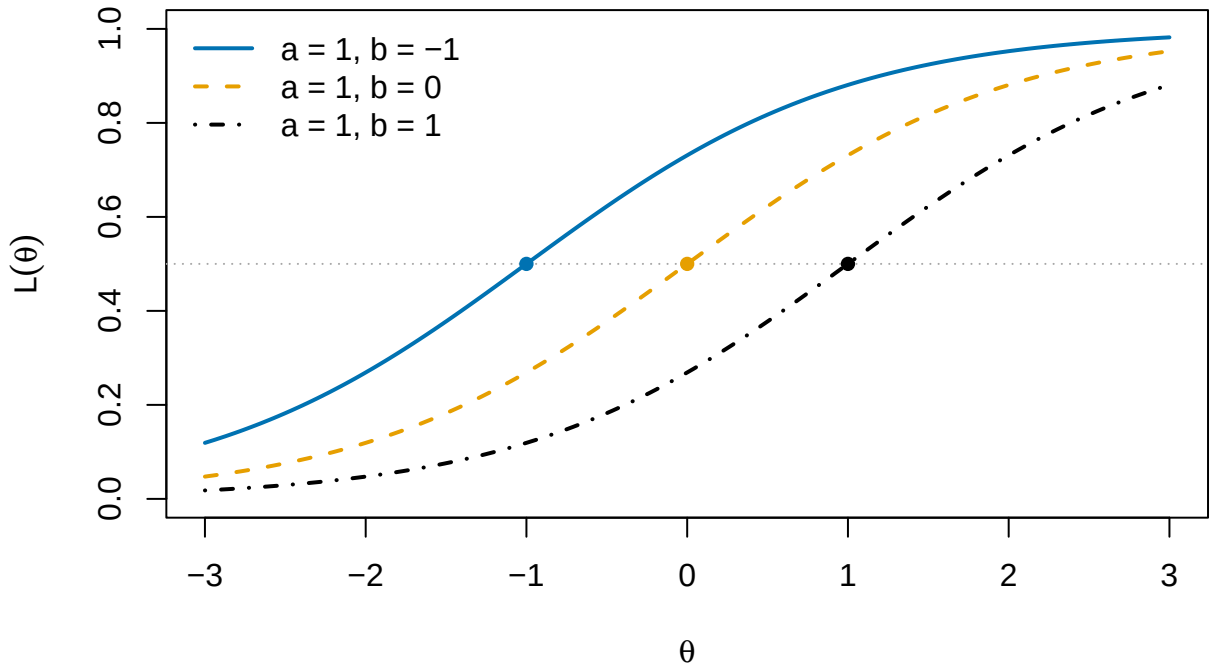
3.5 Logistic Location Curves

$$L(\theta; a, b^*) = \text{logit}^{-1}\{a(\theta - b^*)\}, \qquad b^* = -b_j/a_j.$$

$$\frac{\partial L}{\partial \theta} = aL(1 - L), \qquad \left.\frac{\partial L}{\partial \theta}\right|_{\theta = b^*} = \frac{a}{4}.$$

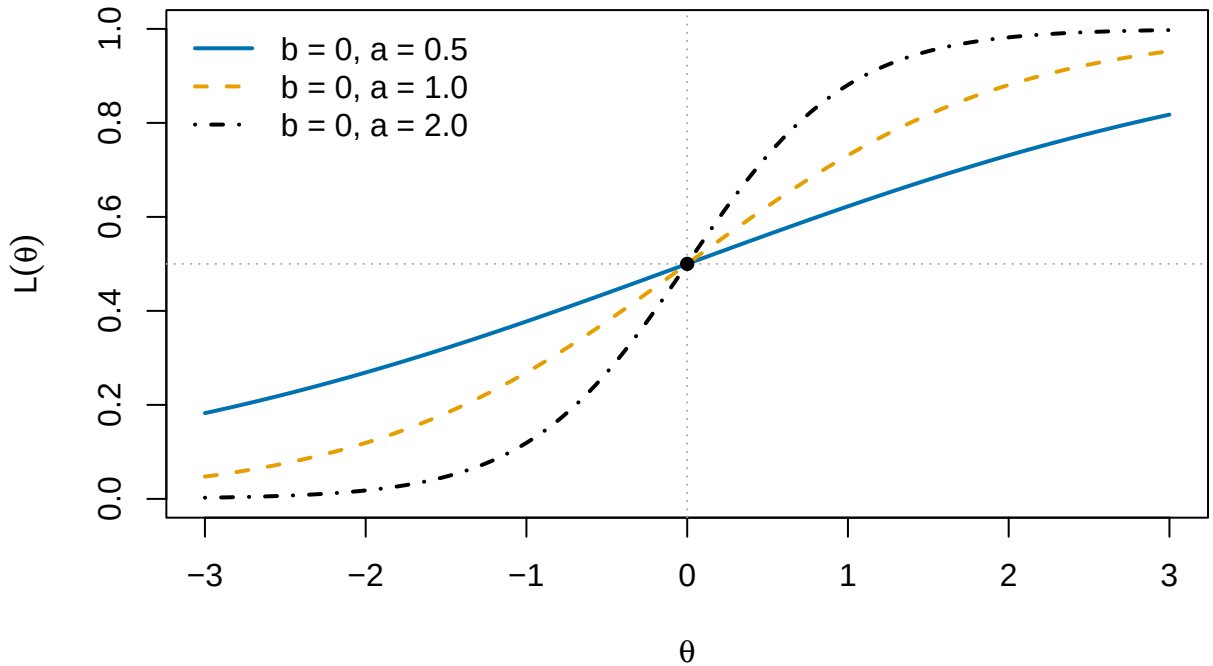
- For beta and simplex, this is the interior conditional mean.
- For Johnson S_B, it is the interior conditional median.
- Difficulty determines the half-response position; discrimination determines the midpoint slope.
- With inflation, these are interior location curves, not the overall response mean.

3.6 Fixed Discrimination, Varying Difficulty



The figure uses b for difficulty b^*; discrimination is fixed at $a = 1$.

3.7 Fixed Difficulty, Varying Discrimination



Difficulty is fixed at $b = 0$; midpoint slopes are 0.125, 0.25, and 0.50.

4. Zero and One Inflation

Let $s(x) = \text{logit}^{-1}(x)$ and $\gamma_{0j} < \gamma_{1j}$:

$$
\begin{aligned}
p_{0ij} &= s(\gamma_{0j} - a_j\theta_i), \\
p_{Iij} &= s(\gamma_{1j} - a_j\theta_i) - s(\gamma_{0j} - a_j\theta_i), \\
p_{1ij} &= 1 - s(\gamma_{1j} - a_j\theta_i).
\end{aligned}
$$

- The probabilities sum to one.
- Zero probability decreases as the trait increases.
- One probability increases as the trait increases.
- Endpoints contribute information about the latent trait (Molenaar et al. 2022).

4.1 Mixed Response Distribution

$$p(y \mid \theta_i, \boldsymbol{\tau}_j) = \begin{cases} p_{0ij}, & y = 0, \\ p_{Iij} f(y \mid \eta_{ij}, \zeta_j), & 0 < y < 1, \\ p_{1ij}, & y = 1. \end{cases}$$

The overall conditional mean is

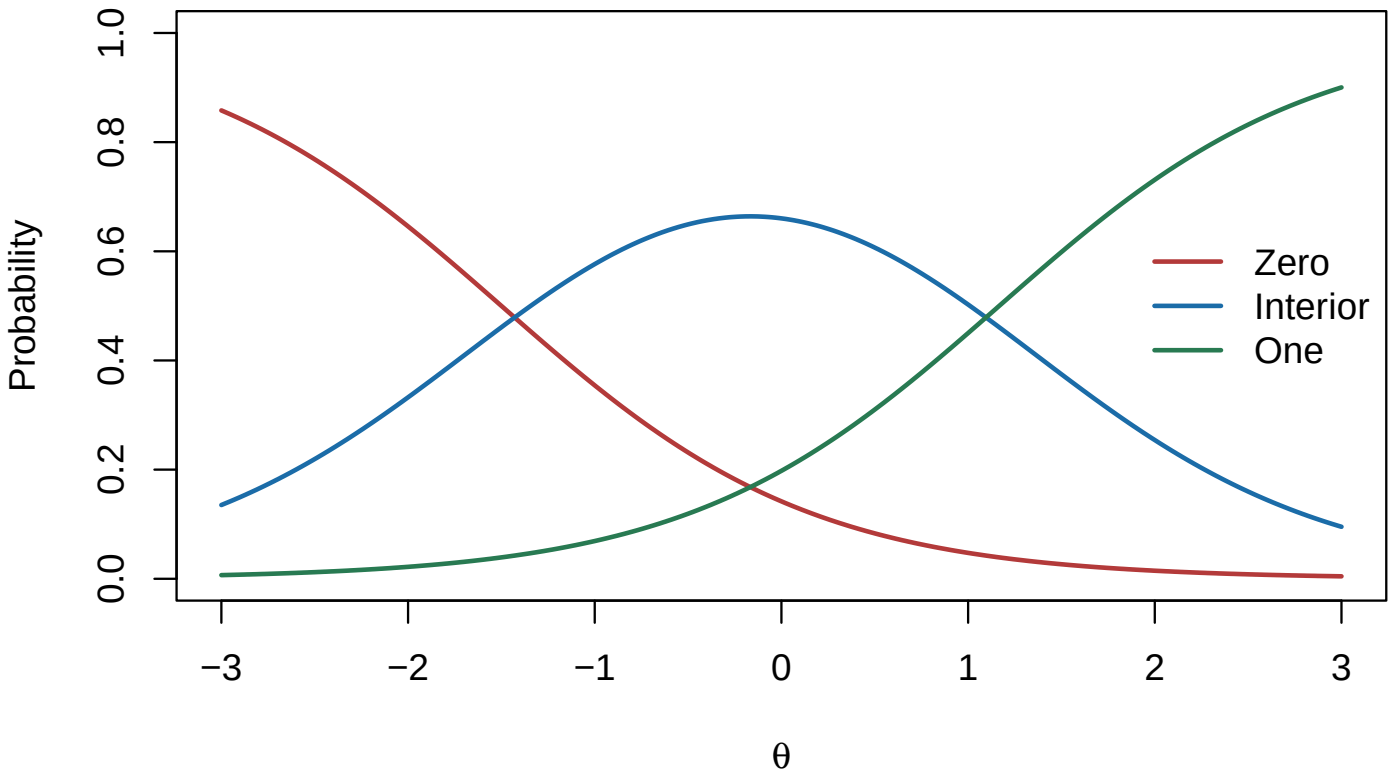
$$E(Y_{ij} \mid \theta_i) = p_{1ij} + p_{Iij} m_{Iij},$$

where m_{Iij} is the interior conditional mean.

The overall response curve includes both boundary probabilities and the interior distribution.

Ospina and Ferrari (2012) model zero-or-one inflation in beta regression; here the boundary probabilities depend on the latent trait (Molenaar et al. 2022).

4.2 Boundary Probability Curves



4.3 Inflation Structure in BoundIRT

Observed endpoints	Admissible inflation rule
Neither endpoint	`"none"` or `"auto"`
Zeros only	`"zero"` or `"auto"`
Ones only	`"one"` or `"auto"`
Both endpoints	`"both"` or `"auto"`

- `inflated = "auto"` includes the endpoints observed in each item (Molenaar 2026).
- Inspect zeros, ones, missingness, and interior responses item by item.
- Arbitrary replacement of endpoints can alter parameter estimates.

5. Bayesian Estimation

Under local independence, the posterior is proportional to

$$\left\{ \prod_i \prod_{j \in \mathcal{O}_i} p(y_{ij} \mid \theta_i, \tau_j) \right\} \left\{ \prod_i g(\theta_i) \right\} \left\{ \prod_j h(\tau_j) \right\}.$$

- $\mathcal{O}_i$ indexes the observed items for person i.
- Fix $\theta_i \sim N(0, 1)$ and constrain $a_j > 0$.
- Sample positive item parameters on logarithmic scales.
- Stan enforces the ordering of boundary parameters.
- `BoundIRT::fitBIRT()` supplies the implementation (Molenaar 2026).

5.1 Prior Distributions

Parameter	Default sampling-scale prior
θ_i	$N(0, 1)$
$\log(a_j)$	$N(0, 10^2)$
b_j	$N(0, 10^2)$
$\log(\zeta_j)$	$N(0, 10^2)$
γ_{0j}, γ_{1j}	$N(0, 10^2)$ with ordering when both occur

- ζ_j is δ_j (Samejima), λ_j (beta), ϕ_j (simplex), or σ_j^2 (normal).
- The package calls this family-specific parameter `phi` (Molenaar 2026).
- Examine sensitivity to the broad item priors, especially for Bayes factors.

5.2 Multiple-Chain Estimation

```
fit <- BoundIRT::fitBIRT(
  z = response_matrix,
  model = "beta", inflated = "auto",
  iter = 4000, warmup = 2000, nchains = 4,
  prior = list(sA = 2, sB = 5, sP = 3, sG = 5),
  seed = 2022,
  control = list(adapt_delta = 0.95,
                 max_treedepth = 12)
)
```

Check $\widehat{R}$, bulk and tail effective sample sizes, trace plots, divergences, and treedepth warnings before interpretation.

5.3 Posterior Predictive Checking

```
replicated <- BoundIRT::repBIRT(fit, nrep = 500)
y_rep <- replicated$z_rep
```

Compare observed and replicated:

- itemwise zero and one proportions,
- interior means, variances, quantiles, and density shapes,
- response frequencies near the boundaries,
- person-level total scores.

Sampling diagnostics assess posterior exploration. Predictive checks assess whether the model reproduces the data.

5.4 Person-Marginal Predictive Criteria

$$p(\mathbf{y}_i \mid \mathbf{T}) = \int \prod_{j \in \mathcal{O}_i} p(y_{ij} \mid \theta, \boldsymbol{\tau}_j) \varphi(\theta)\, d\theta.$$

```
log_lik <- BoundIRT::loglikBIRT(
  fit, marginal = TRUE, nquad = 15)
loo_result <- loo::loo(log_lik)
waic_result <- loo::waic(log_lik)
```

- Person-marginal criteria evaluate prediction for new persons.
- Smaller LOOIC and WAIC indicate better predictive fit on the same data.
- Examine difference uncertainty and Pareto-k diagnostics (Vehtari et al. 2017).

5.5 Marginal Likelihood and Bayes Factors

$$p(\mathbf{Y} \mid M) = \int\int p(\mathbf{Y} \mid \boldsymbol{\theta}, \mathbf{T}, M) p(\boldsymbol{\theta} \mid M) p(\mathbf{T} \mid M)\, d\boldsymbol{\theta}\, d\mathbf{T}.$$

$$\log BF_{12} = \log p(\mathbf{Y} \mid M_1) - \log p(\mathbf{Y} \mid M_2).$$

- `bridgeBIRT()` estimates log marginal likelihood by bridge sampling (Gronau et al. 2017; Molenaar 2026).
- Larger log marginal likelihood favors a model.
- Use defensible priors and examine prior sensitivity.
- Integrating over item parameters distinguishes this quantity from the person-marginal likelihood.

6. Abasement Scale: Data and Context

- Continuous-response version of the Adjective Check List Abasement scale (Gough and Heilbrun 1980; Molenaar et al. 2022).
- 244 people responded to 10 adjectives on 60-millimeter line segments.
- Responses rescale to $[0, 1]$.
- `self-confident*` is already reverse scored in the package data.
- Of 2,440 item responses, approximately 10.74% are zeros. There are no exact ones.
- Compare three zero-inflated interior families using the saved fits.

6.1 Installation of BoundIRT

The CRAN package is `BoundIRT`; the manuscript uses version 0.6.0 (Molenaar 2026).

```
# Install once; do not reinstall during rendering.
install.packages("BoundIRT",
                 repos = "https://cloud.r-project.org")
library(BoundIRT)
packageVersion("BoundIRT")
```

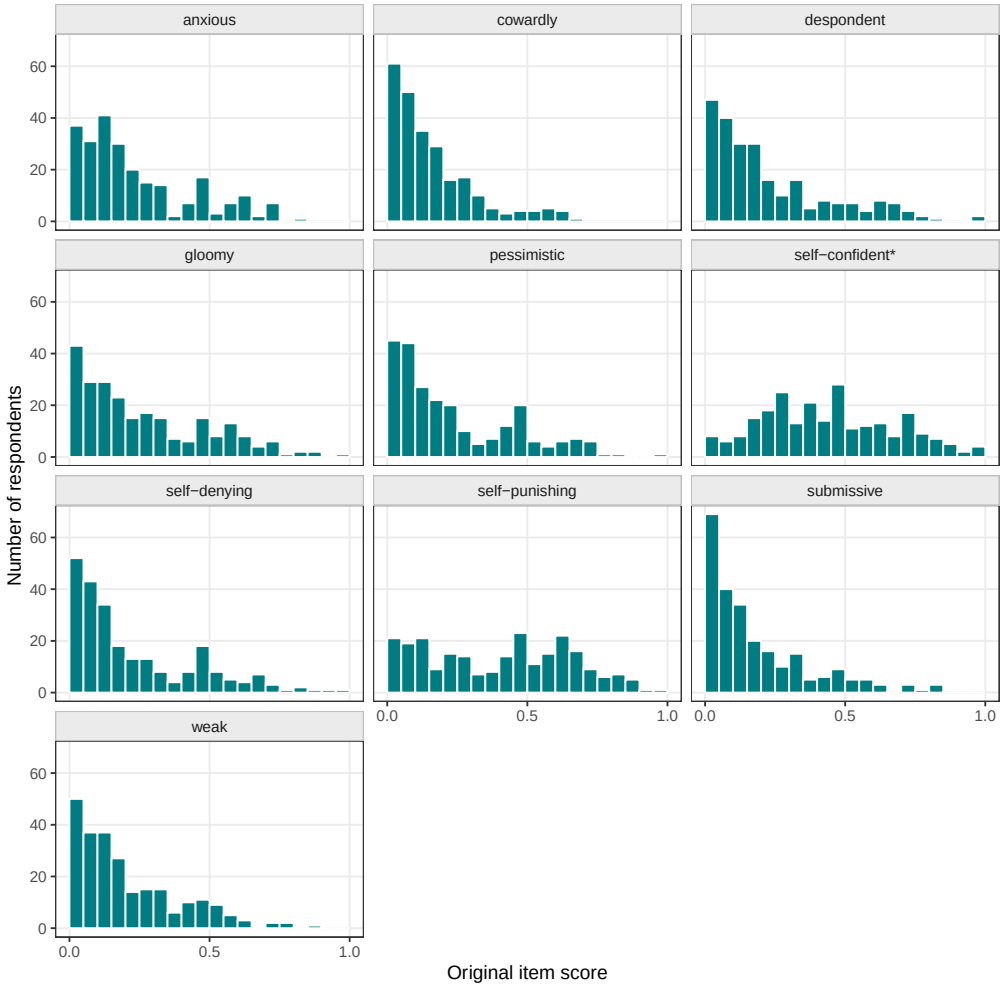
- RStan and compiled Stan support require a working C++ toolchain.
- The manuscript also supports a project-local `Rlib` library.
- Its Abasement analysis uses distributed or saved fits, without rerunning MCMC during rendering.

6.2 BoundIRT Functions

Function	Purpose
`fitBIRT()`	Fit Samejima, beta, simplex, or normal bounded IRT models
`rBIRT()`	Simulate bounded IRT data, with optional zero/one inflation
`coef()`	Extract posterior means and optional posterior standard deviations
`plot()`	Plot ICCs, item information, test information, or fitted densities
`repBIRT()`	Generate posterior predictive response datasets
`loglikBIRT()`	Compute person-marginal or conditional log likelihoods
`bridgeBIRT()`	Estimate the log marginal likelihood by bridge sampling
`latregBIRT()`	Regress the latent person variable on observed predictors

Version 0.6.0 (Molenaar 2026).

6.3 Original Item Distributions



6.4 Boundary Use and Item Analysis

Item	Zeros	Zero proportion	Corrected item–total r
anxious	26	0.107	0.372
cowardly	39	0.160	0.405
despondent	27	0.111	0.611
submissive	40	0.164	0.328
pessimistic	30	0.123	0.546
gloomy	29	0.119	0.649
self-punishing	10	0.041	0.269
self-denying	30	0.123	0.505
self-confident*	2	0.008	0.332
weak	29	0.119	0.452

6.5 Fitted Zero-Inflated Models

For every item,

$$p_{0ij} = \text{logit}^{-1}(\gamma_{0j} - a_j\theta_i), \qquad p_{Iij} = 1 - p_{0ij}.$$

The interior follows a beta, Johnson S_B, or simplex distribution.

```
library(BoundIRT)
data(Abasement)
# Multiple chains for a new analysis.
fit_jsb <- fitBIRT(z = Abasement, model = "samejima",
                   inflated = "zero", nchains = 4)
```

The source uses distributed or saved fits. The Johnson S_B and simplex examples use one chain with 2,000 iterations, including 1,000 warmup iterations.

6.6 Model Comparison: Original Responses

Person-marginal predictive criteria:

Interior family	LOOIC	SE	WAIC	SE
Beta	-517.98	168.23	-517.93	168.29
Johnson S_B	-675.85	165.33	-676.20	165.31
Simplex	-521.52	185.17	-521.94	185.09

- Johnson S_B has the smallest reported criteria.
- Retain it for the illustrative item and person interpretations.
- WAIC flags influential observations. Simplex has one Pareto-$k > 1$ case.
- Single-chain examples require confirmation with multiple chains.

6.7 Sensitivity to Endpoint Transformation

The source also fits non-inflated models after transforming every response:

$$Y_{ij}^{\text{new}} = \frac{Y_{ij}(n-1) + 0.5}{n}, \qquad n = 244.$$

Interior family	LOOIC	WAIC
Beta	-3031.09	-3031.26
Johnson S_B	-2770.90	-2771.19
Simplex	-793.71	-781.70

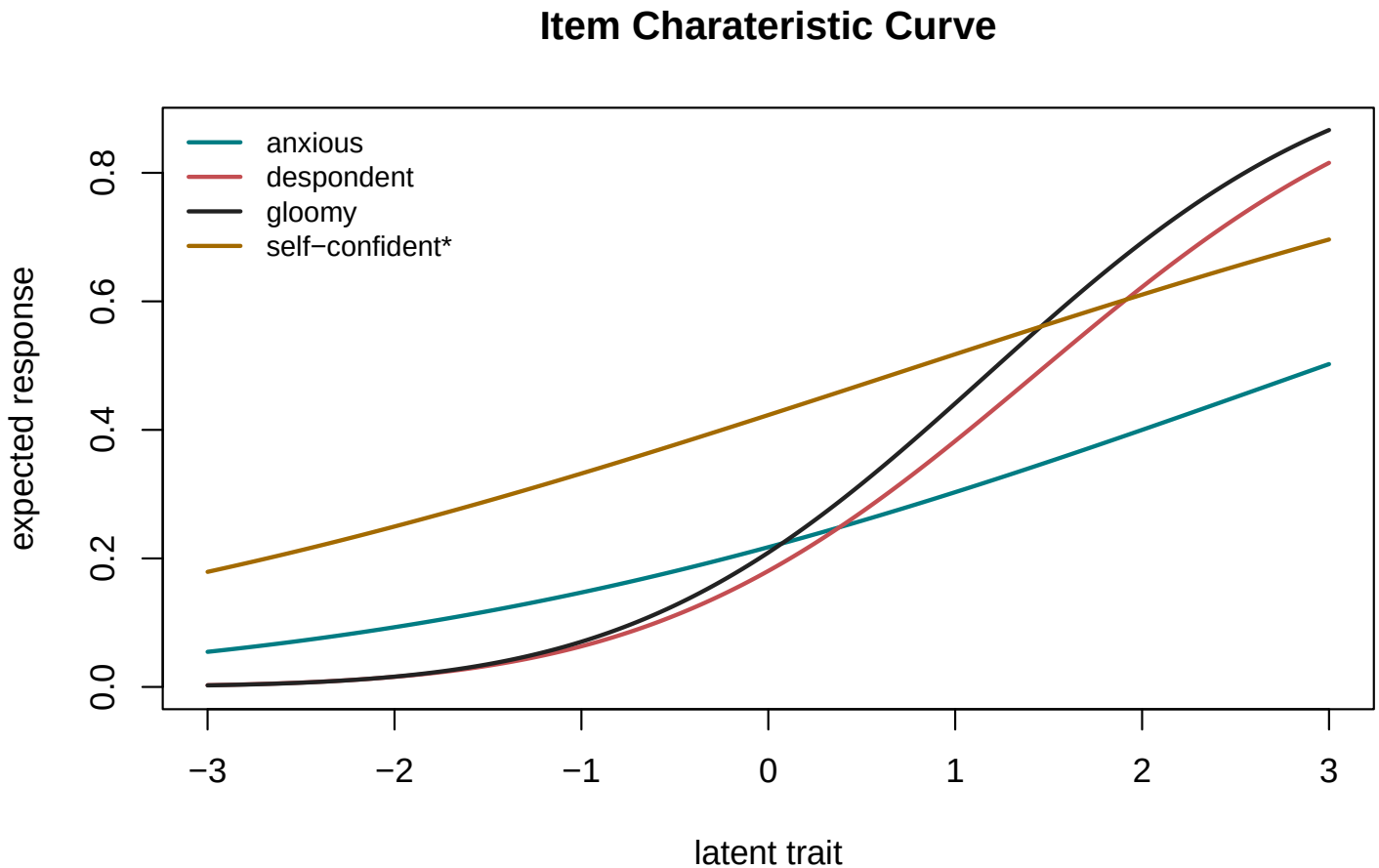
- Beta has the smallest criteria within this transformed-data group.
- Compare criteria only within a common response dataset.
- Transformed simplex is provisional: two Pareto-$k > 1$ cases, maximum $\widehat{R} = 1.027$, minimum ESS 55.

6.8 Selected Model: Item Parameters

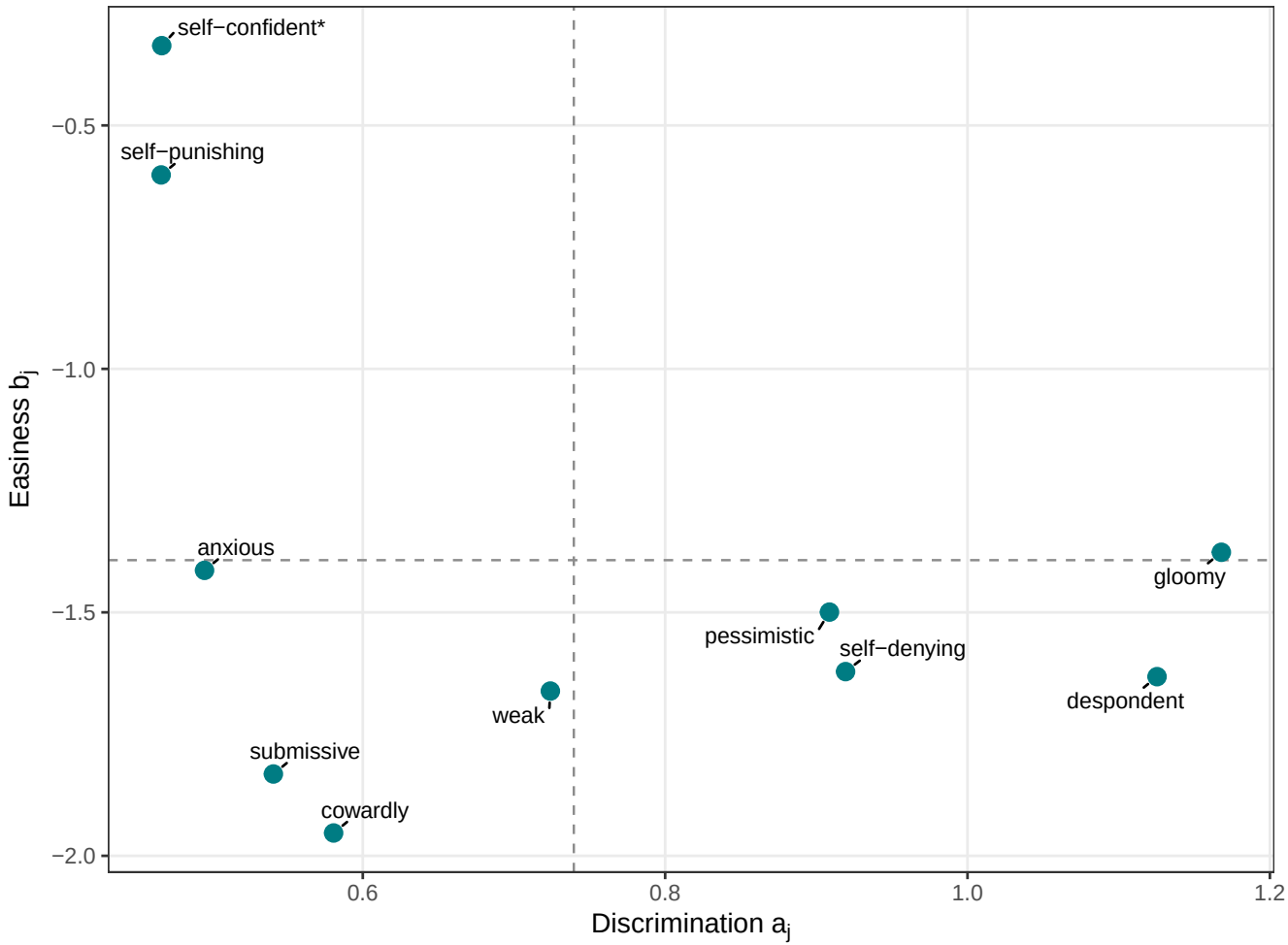
Posterior means for the zero-inflated Johnson S_B model:

Item	a_j	b_j	δ_j	γ_{0j}
anxious	0.495	-1.414	1.204	-2.238
cowardly	0.581	-1.953	0.912	-1.675
despondent	1.125	-1.632	0.841	-2.387
submissive	0.541	-1.832	1.460	-1.632
pessimistic	0.909	-1.499	1.043	-2.170
gloomy	1.168	-1.377	0.680	-2.321
self-punishing	0.467	-0.602	1.845	-3.015
self-denying	0.919	-1.622	1.374	-2.191
self-confident*	0.467	-0.336	1.296	-3.682
weak	0.724	-1.662	1.017	-2.169

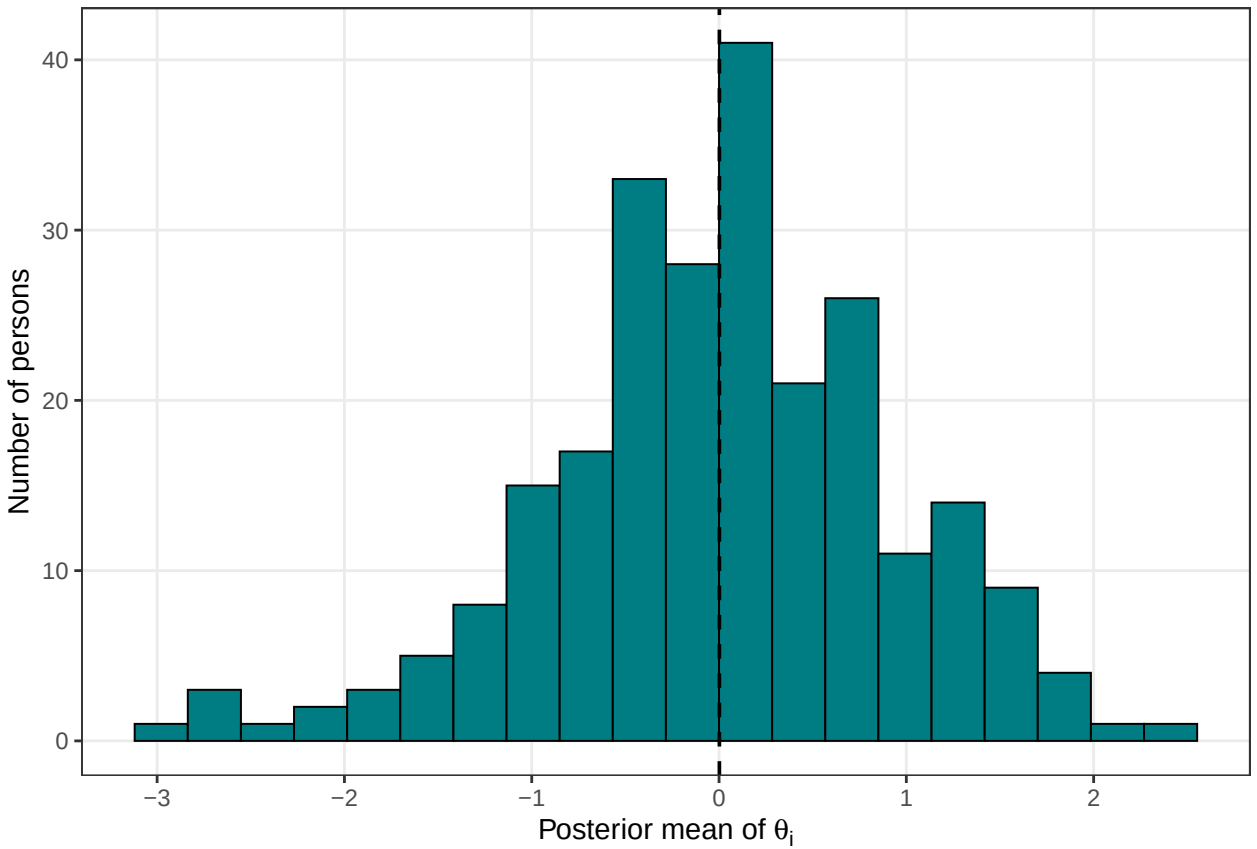
6.9 Selected Model: Item Characteristic Curves



6.10 Selected Model: Discrimination and Easiness

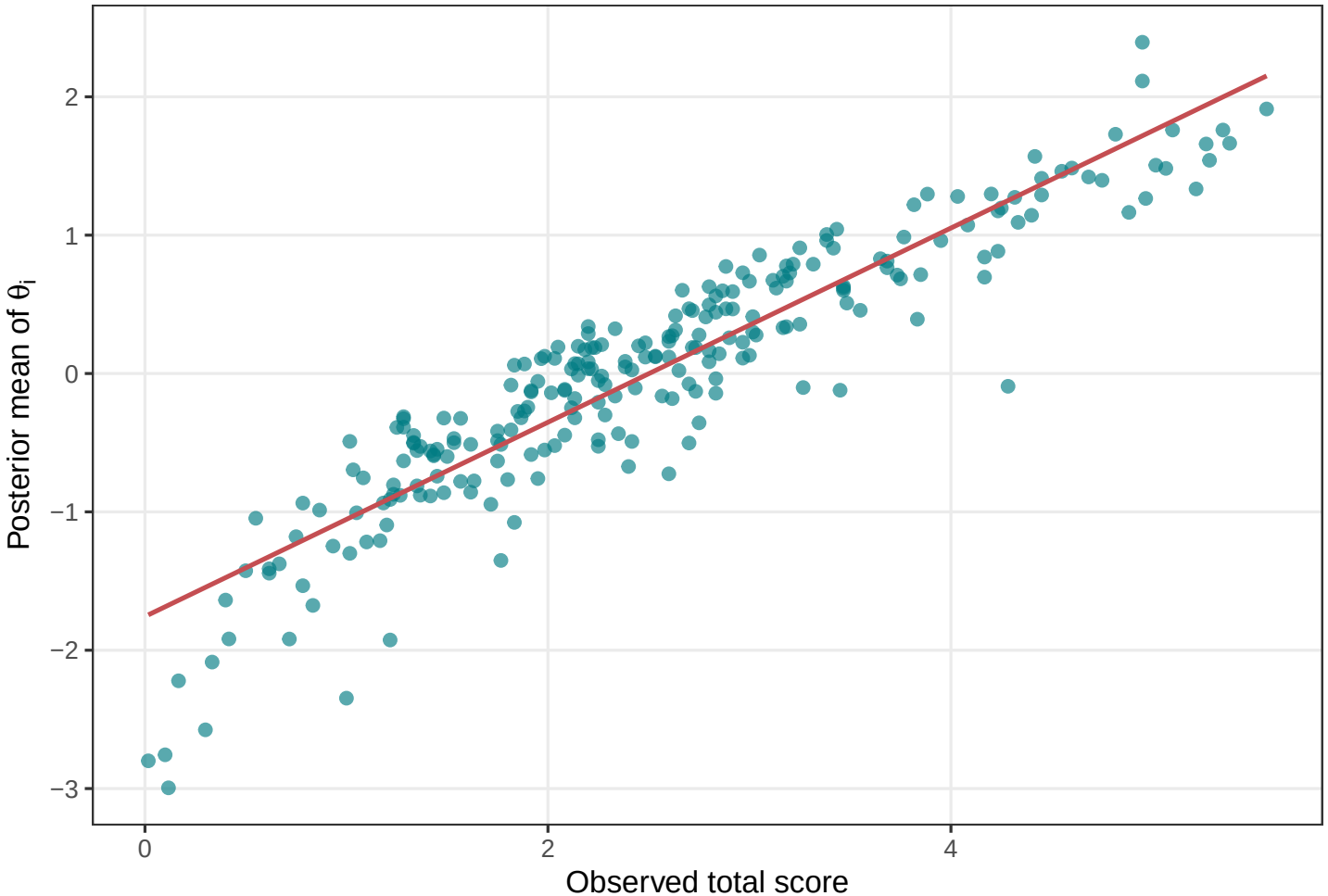


6.11 Selected Model: Person Estimates



244 posterior means: mean 0.002, SD 0.922, range $[-2.995, 2.394]$.

6.12 Total Score and Estimated Trait

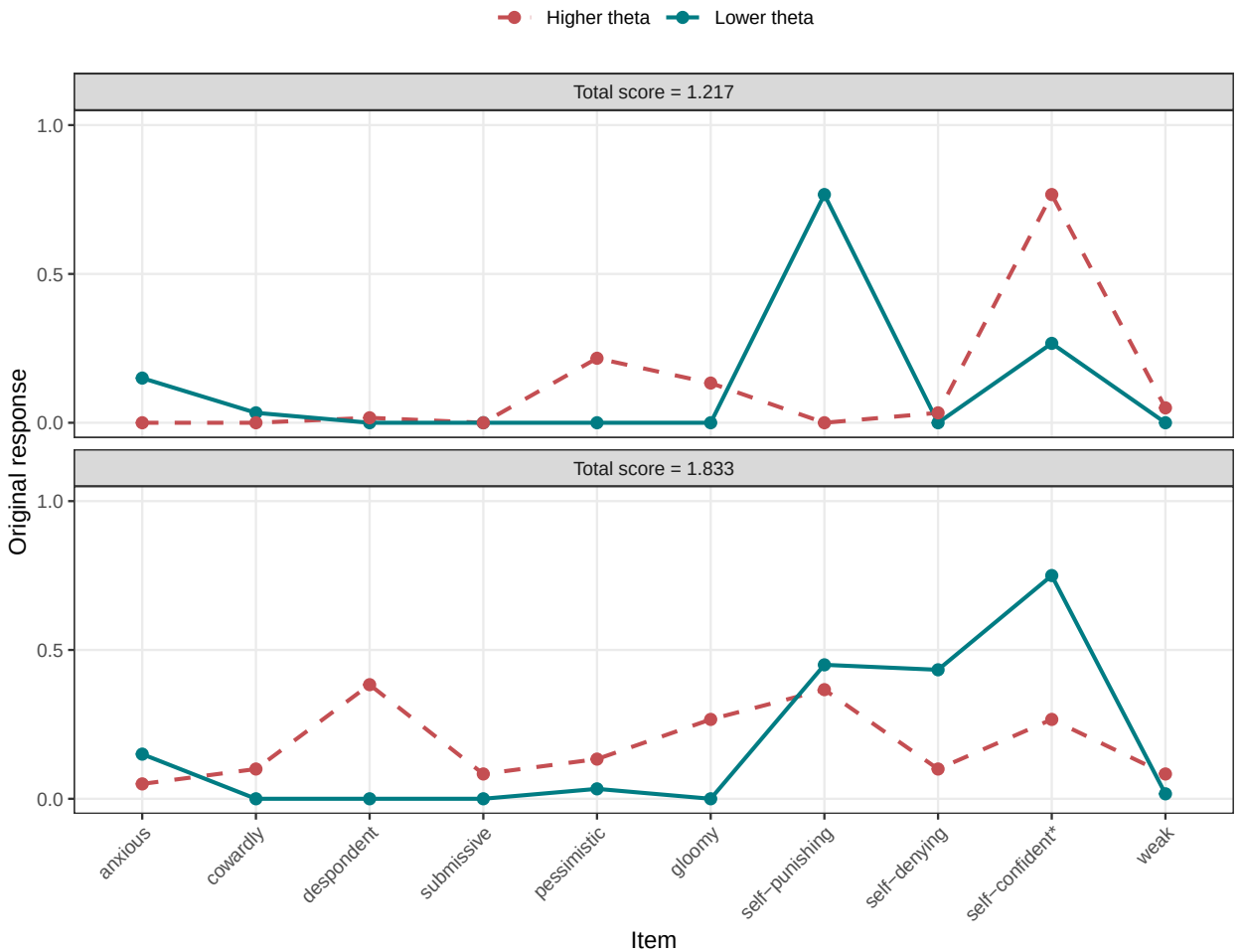


6.13 Equal Scores and Different Trait Estimates

Person	Total score	Posterior mean θ	Posterior SD
238	1.833	-1.076	0.489
59	1.833	0.060	0.363
94	1.217	-1.926	0.592
46	1.217	-0.910	0.367

- The total score sums the ten original responses without item weights.
- IRT uses the item-response pattern and estimated item parameters.
- These pairs come from the tied-score groups with the largest trait-estimate spreads in the source analysis.
- Posterior uncertainty remains relevant when interpreting differences.

6.14 Equal-Score Response Profiles



7. Practical Analysis Workflow

1. Rescale using theoretical bounds and inspect endpoint use by item.
2. Fit plausible interior families with an appropriate inflation structure.
3. Check multiple-chain sampling diagnostics and prior sensitivity.
4. Evaluate replicated endpoints, interior distributions, and total scores.
5. Compare person-marginal predictive criteria on the same response data.
6. Interpret item curves and person estimates with uncertainty.
7. Examine dimensionality and local dependence before substantive conclusions.

8. Exercises: Estimation and Simulation

1. Fit Samejima's model to Abasement using `EstCRM` (Samejima 1973; Zopluoglu 2022). Align the latent scale and parameterizations, and explain endpoint handling.
2. Analyze the 307-person, 10-item `SelfEff` dataset in `EstCRM` (Zopluoglu 2022). Use theoretical bounds, inspect endpoints, and fit the three bounded IRT families.
3. Simulate each bounded IRT family, fit all three candidates, and compare parameter and distributional recovery.
4. Investigate whether the model favored by predictive criteria also best recovers person parameters. Report uncertainty and convergence diagnostics.

8.1 Exercises: Applications and Methods

5. Analyze the municipal economy, education, and health indices in Oliveira et al. (2026). Fit the bounded IRT families and assess whether a single latent dimension is appropriate.
6. Assess dimensionality and local dependence in Abasement and the exercise datasets. Explain their effect on interpretation.
7. Implement marginal likelihood estimation of item parameters, integrating over the person distribution.

9. Conclusions

- Respect the theoretical response bounds and inspect exact endpoints.
- Johnson S_B, beta, and simplex models imply different variance and information structures.
- The inflation framework lets endpoint responses contribute to measurement.
- The Abasement example favors Johnson S_B among the original-data fits, subject to sampling and predictive-diagnostic limitations.
- Combine predictive comparison with model checks and substantive interpretation.
- `BoundedIRT.Rmd` contains the full derivations and analyses.

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